

João Caldeira

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Education

Feb 2023 – Nov 2024 ♦ **M.Sc. COMPUTER SCIENCE AND ENGINEERING @ Instituto Superior Técnico, Universidade de Lisboa**

Sep 2018 – Feb 2023 ♦ **B.Sc. COMPUTER SCIENCE AND ENGINEERING @ Instituto Superior Técnico, Universidade de Lisboa**

Professional Experience

Mar 2025 – Present ♦ **SOFTWARE ENGINEER @ Infosistema**

- ▶ **Founding engineer and Tech Lead** of bizSupply, an AI-powered contract-intelligence SaaS: took it from an empty repo to production across 18 microservices, a Chrome extension, and a public Python SDK.
- ▶ Architected the distributed, event-driven **microservices** platform on **GCP** powering an OCR- and LLM-based document pipeline: priority-queued Redis dispatch, noisy-neighbor-safe scheduling, heartbeat-based crash recovery, and a queue-depth autoscaler.
- ▶ Built a provider-agnostic **LLM orchestration** platform (Vertex Gemini, Claude, OpenAI, open models) with multi-model routing, fallback, and context caching; cost-tier routing targeted a 30% spend cut.
- ▶ Built an **MCP server** exposing ~50 tenant-scoped tools across the platform (documents, pipelines, ontologies, jobs) to Claude Desktop, Cursor, and an in-app assistant, with OAuth 2.1 JWT-secured tool access.
- ▶ Owned Infrastructure-as-Code and CI/CD end-to-end: 22 **Terraform** modules across 5 environments, 42 GitHub Actions workflows, **Workload Identity Federation**, and policy-as-code gates; cut observability spend ~87% and compute costs ~40%.
- ▶ Designed the storage layer for a multi-tenant platform: a three-database topology (operational, analytics, auth) with **Alembic**-managed migrations, custom **asyncpg** data-access mixins replacing a traditional ORM, and a medallion schema with a dedicated no-PII tier.
- ▶ Created and grew the company's **Claude Code** plugin ecosystem: authored a docs-scaffolding, git-worktree, and milestone-automation plugin, consolidated into a company-wide marketplace and adopted org-wide.
- ▶ Built a natural-language-to-DSL search engine with **RAG** and a synthetic-data pipeline that fine-tuned an OpenAI model, shipped to production with improved accuracy and lower inference cost versus the base model.

Jul 2023 – Mar 2025 ♦ **QA AUTOMATION ENGINEER @ SKY Portugal**

- ▶ Promoted from summer intern to full-time QA Automation Engineer after building automated resilience-testing frameworks spanning mobile, TV, and web platforms.
- ▶ Engineered **API telemetry** and **CI/CD** improvements that cut defect rates and raised test-suite adoption and confidence across client teams.

Publications

- 2026 ♦ **Dynamic Ensemble of Specialized Models for Multi-Timeframe Stock Market Trend Prediction**
João Caldeira, Rui Neves – *Computational Economics* (Springer) – 10.1007/s10614-025-11252-0
Peer-reviewed, profitable algorithmic trading strategy that outperformed the S&P 500 and NASDAQ-100. Built on a **novel data-labeling method** that separates short- and long-term market trends, feeding a **specialized ensemble of prediction models** trained per horizon. Originated from MSc thesis research.

Skills

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|---------------------|---|
| Languages | ♦ Python · C# · Java · JavaScript · TypeScript · C++ · SQL |
| Technologies | ♦ .NET · FastAPI · Spring · Next.js · React · PostgreSQL · Redis · Docker · Kubernetes · Terraform · Google Cloud · AWS · GitHub Actions · Vercel · Grafana · PyTorch · LLMs · RAG · MCPs · Agent Development Kit · Claude Code · Playwright · PyTest |
| Soft Skills | ♦ Critical Thinking · Problem-Solving · Communication · Collaboration · Team Spirit |